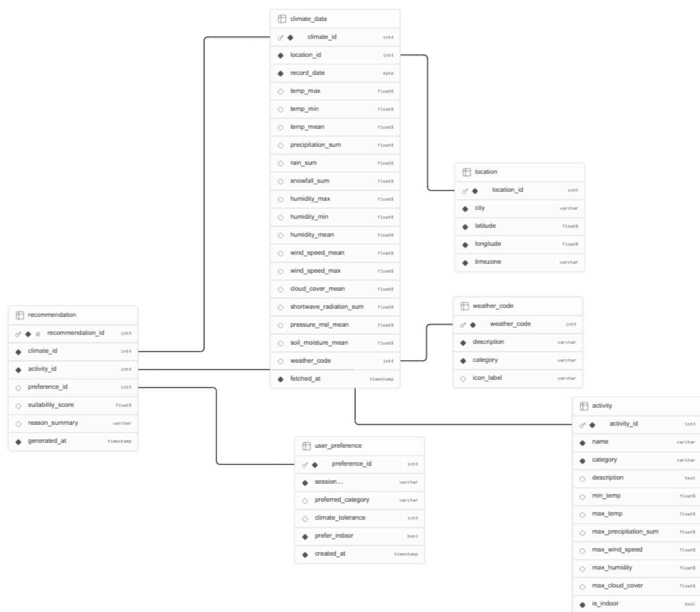
Database Schema Documentation

New Orleans Climate Aware Recommender

ER Diagram



Database Overview

This database stores climate data and activity recommendation information loaded into a PostgreSQL database hosted in Supabase. The schema supports matching daily climate conditions to suitable outdoor and indoor activities based on user preferences in New Orleans, Louisiana.

The schema is normalized to approximately Third Normal Form (3NF) by:

- Separating entities into related tables
- Avoiding repeated data across records
- Using foreign keys to express relationships between tables

The database contains six tables:

Table	Purpose
location	Stores geographic coordinates and timezone information for the New Orleans forecast location

Table	Purpose
weather_code	Stores standardized WMO weather condition codes and descriptions
climate_data	Stores daily climate records fetched from the Open-Meteo Climate API
user_preference	Stores session-level user preferences used to filter activity recommendations
activity	Stores activity details and weather suitability thresholds
recommendation	Stores generated activity recommendations linked to climate data and activities

Table Documentation

1. location

Purpose: Stores unique geographic location information used as the basis for climate data retrieval. New Orleans, LA (29.9547°N, 90.0751°W) is the primary location stored in this table.

Primary Key: location_id

Relationships:

- One location can have many climate_data records.
- Referenced by climate_data.location_id.

Column Name	Data Type	Key	Description
location_id	SERIAL	**Primary Key**	Auto-generated unique location identifier
city	VARCHAR(100)		Name of the city (e.g. New Orleans)
latitude	FLOAT		Geographic latitude coordinate
longitude	FLOAT		Geographic longitude coordinate
timezone	VARCHAR(50)		IANA timezone string (e.g. America/Chicago)

2. weather_code

Purpose: Stores standardized WMO (World Meteorological Organization) weather interpretation codes along with human-readable descriptions, category groupings, and icon labels.

Primary Key: weather_code

Examples:

- Code 0 = Clear sky
- Code 61 = Rain: Slight intensity

Relationships:

- One weather_code can relate to many climate_data records.
- Referenced by climate_data.weather_code.

Column Name	Data Type	Key	Description
weather_code	INTEGER	**Primary Key**	Unique WMO weather condition code
description	VARCHAR(200)		Human-readable description of the weather condition
category	VARCHAR(100)		Grouped category label (e.g. Clear, Rain, Snow)
icon_label	VARCHAR(50)		Icon identifier for UI display

3. climate_data

Purpose: Stores daily climate records fetched from the Open-Meteo Climate API for New Orleans. This table replaces short-term forecast data with long-term climate variables suitable for activity recommendation and trend analysis.

Primary Key: climate_id

Climate data includes:

- Temperature metrics (max, min, mean)
- Precipitation sum, rain sum, and snowfall sum
- Humidity metrics (max, min, mean)
- Wind speed (mean and max)
- Cloud cover, shortwave radiation, sea level pressure, and soil moisture
- WMO weather condition code

Foreign Keys:

- location_id → location.location_id
- weather_code → weather_code.weather_code

Relationships:

- Many climate_data records can reference one location.
- Many climate_data records can reference one weather code.
- One climate_data record can generate many recommendations.

Column Name	Data Type	Key	Description
climate_id	SERIAL	**Primary Key**	Auto-generated climate record ID
location_id	INTEGER	**Foreign Key**	Link to the location table
record_date	DATE		Date of the climate record
temp_max	FLOAT8		Maximum daily temperature (°C)
temp_min	FLOAT8		Minimum daily temperature (°C)
temp_mean	FLOAT8		Mean daily temperature (°C)
precipitation_sum	FLOAT8		Total daily precipitation (mm)
rain_sum	FLOAT8		Total daily rainfall (mm)
snowfall_sum	FLOAT8		Total daily snowfall (mm)
humidity_max	FLOAT8		Maximum relative humidity (%)
humidity_min	FLOAT8		Minimum relative humidity (%)
humidity_mean	FLOAT8		Mean relative humidity (%)
wind_speed_mean	FLOAT8		Mean wind speed at 10m (km/h)
wind_speed_max	FLOAT8		Maximum wind speed at 10m (km/h)
cloud_cover_mean	FLOAT8		Mean cloud cover (%)
shortwave_radiation_sum	FLOAT8		Total shortwave solar radiation (MJ/m^2)
pressure_msl_mean	FLOAT8		Mean sea level pressure (hPa)
soil_moisture_mean	FLOAT8		Mean soil moisture 0–10 cm (m^3/m^3)
weather_code	INTEGER	**Foreign Key**	WMO weather condition code for the day
fetched_at	TIMESTAMP		Timestamp when the climate data was fetched

4. user_preference

Purpose: Stores session-level user preferences that are used to filter and personalize activity recommendations. Each record represents one user session's configuration.

Primary Key: preference_id

Example:

- Session prefers Outdoor Sports category, climate_tolerance = 3, prefer_indoor = false

Relationships:

- One user_preference record filters many recommendation results.
- Referenced by recommendation.preference_id.

Column Name	Data Type	Key	Description
preference_id	SERIAL	**Primary Key**	Auto-generated preference identifier
session_id	VARCHAR(100)		Session or user identifier
preferred_category	VARCHAR(100)		User's preferred activity category
climate_tolerance	INTEGER		Tolerance for extreme climate conditions (scale value)
prefer_indoor	BOOLEAN		Whether the user prefers indoor activities (default: false)
created_at	TIMESTAMP		Record creation timestamp (default: NOW())

5. activity

Purpose: Stores New Orleans tourism and leisure activity information along with climate suitability thresholds. These thresholds are used to match activities to daily climate conditions.

Primary Key: activity_id

Activity categories (from proposal):

- Outdoor Sports — hiking, biking, kayaking along the Mississippi
- Indoor Activities — museums, shopping, jazz venues, dining
- Weather-Dependent Events — street parades, crawfish boils, outdoor festivals
- Seasonal Activities — fall cultural events, holiday season activities

Relationships:

- One activity can be included in many recommendation records.

- Referenced by recommendation.activity_id.

Column Name	Data Type	Key	Description
activity_id	SERIAL	**Primary Key**	Auto-generated activity identifier
name	VARCHAR(100)		Name of the activity
category	VARCHAR(100)		Activity category (e.g. Outdoor Sports, Indoor Activities)
description	TEXT		Detailed description of the activity
min_temp	FLOAT8		Minimum suitable temperature (°C)
max_temp	FLOAT8		Maximum suitable temperature (°C)
max_precipitation_sum	FLOAT8		Maximum tolerable precipitation sum (mm)
max_wind_speed	FLOAT8		Maximum tolerable wind speed (km/h)
max_humidity	FLOAT8		Maximum tolerable humidity (%)
max_cloud_cover	FLOAT8		Maximum tolerable cloud cover (%)
is_indoor	BOOLEAN		Whether the activity is indoors (default: false)

6. recommendation

Purpose: Stores generated activity recommendations that link a climate data record to a suitable activity. Each record includes a suitability score and a natural language reason summary, filtered by the user's session preferences.

Primary Key: recommendation_id

Foreign Keys:

- climate_id → climate_data.climate_id
- activity_id → activity.activity_id
- preference_id → user_preference.preference_id

Relationships:

- Many recommendations can be generated from one climate_data record.

- Many recommendations can reference one activity.
- Each recommendation is directly linked to the user_preference that filtered it.

Column Name	Data Type	Key	Description
recommendation_id	INTEGER	**Primary Key**	Auto-generated recommendation identifier
climate_id	INTEGER	**Foreign Key**	Link to the climate data record
activity_id	INTEGER	**Foreign Key**	Link to the recommended activity
preference_id	INTEGER	**Foreign Key**	Link to the user preference that filtered this recommendation
suitability_score	FLOAT		Score indicating how well the activity fits the climate (0–1)
reason_summary	VARCHAR(500)		Natural language explanation of the recommendation
generated_at	TIMESTAMP		Timestamp when the recommendation was created

Cardinality Relationships

Parent Table	Child Table	Relationship Type
location	climate_data	One-to-Many
weather_code	climate_data	One-to-Many
climate_data	recommendation	One-to-Many
activity	recommendation	One-to-Many
user_preference	recommendation	One-to-Many

Normalization Notes (3NF)

This schema is normalized to approximately Third Normal Form:

- Weather condition descriptions are separated into weather_code to avoid storing repeated descriptions in every climate_data row.
- Geographic information is isolated in location, allowing multiple climate records to reference the same place without duplication.

- Activity climate thresholds are stored once in activity and reused across many recommendation records.
- User preferences are separated into user_preference and directly referenced by recommendation via preference_id.
- Non-key columns depend only on each table's primary key.

Data Sources

Climate Data Source

Climate data is sourced from the Open-Meteo Climate API. The API endpoint used is:

- Endpoint: https://open-meteo.com/en/docs/climate-api
- Location: New Orleans, LA (29.9547°N, 90.0751°W)

The API provides:

- Daily climate records covering temperature, precipitation, rain, snowfall, humidity, wind speed, cloud cover, shortwave radiation, sea level pressure, and soil moisture
- Weather condition codes following WMO Weather Interpretation standards
- Support for 7 climate models (all selected) for multi-model comparison
- Data freshness tracking via the fetched_at timestamp field

Activity Data Source

Activity data is sourced from a curated dataset of New Orleans tourism attractions and leisure activities containing:

- Activity names, categories, and descriptions specific to New Orleans
- Climate suitability thresholds for each activity type
- Indoor/outdoor classification via the is_indoor boolean field

Example Relationship Flow

The following example illustrates how a recommendation is generated for a New Orleans summer day:

- climate_data.weather_code = 2 (Partly cloudy)
- climate_data.temp_max = 32°C, precipitation_sum = 1.2 mm, humidity_mean = 72%
- user_preference.preferred_category = Outdoor Sports, prefer_indoor = false
- activity.name = Biking, min_temp = 10, max_temp = 32, max_precipitation_sum = 2, max_humidity = 80
- Suitability check passes → recommendation record created with suitability_score = 0.88
- reason_summary = Partly cloudy with low precipitation and acceptable humidity, ideal for a bike ride along the Mississippi

This design ensures that weather descriptions and activity thresholds are never duplicated, and each recommendation is fully traceable back to its source climate record, activity, and user preference.